Setup for Laser Powder Bed Fusion 330

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Class Outline

Laser Bed Powder Fusion
Software
Powder Sieving System
Machine Components
Machine Component Calibration
L-PBF Component and System Review
Software Configuration for Printing
Mirrors and Lenses
Build Surface Setup
Test Prints and Coupons
L-PBF Print Setup Review

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Laser Bed Powder Fusion

Laser powder bed fusion (L-PBF) is an additive manufacturing (AM) technique that builds parts using a laser beam to melt powdered material into layers. Unlike traditional manufacturing, where material properties are uniform throughout a part, L-PBF processes must account for variables in temperature and building strategies that might affect a part's microstructure. Microstructure is the shape and alignment of a material's microscopic components, and it often determines its hardness, toughness, and other properties.

The setup for most L-PBF machines share similar steps, including both automated and manual functions. For industrial L-PBF machines, proper setup helps produce repeatable, consistent results for each production run. Improper setup can create unwanted variations in parts and reduce repeatability, which can increase scrap rates and waste time, energy, and materials. It is therefore crucial to set up each part build correctly.

In L-PBF setup, an operator must actively employ software and understand machine settings, parameters, and atmospheric conditions. Altogether, setup processes for L-PBF can be divided into three general areas: software, the powder sieve system, and the build chamber.

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Figure 1: An L-PBF Machine and Operator

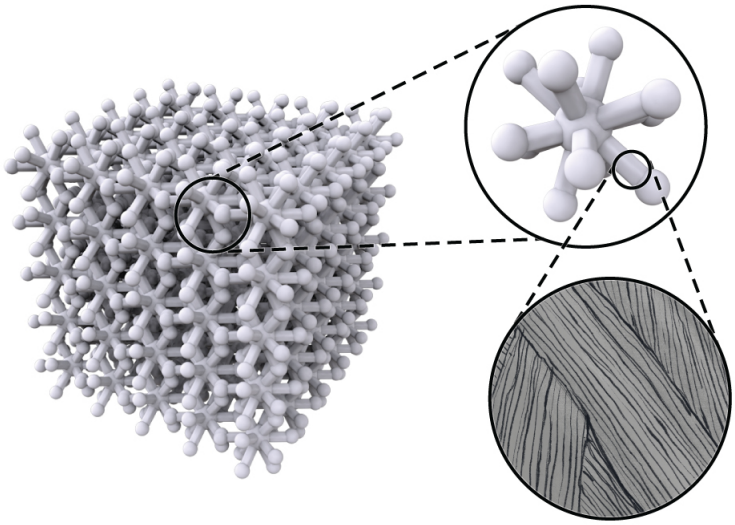

Figure 2: Microstructures can affect L-PBF part building.

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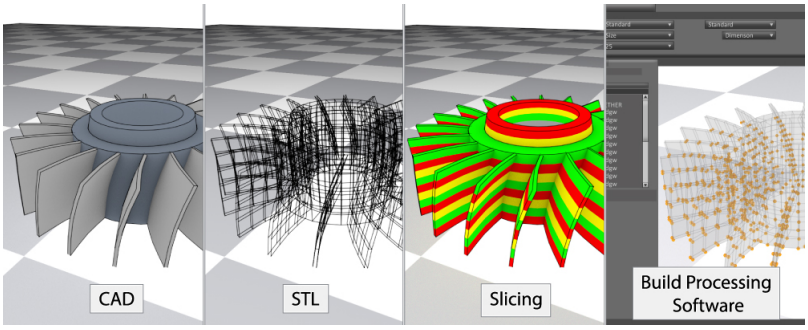
Software



Like other AM methods, engineers using L-PBF work with computer-aided design (CAD) software to create a 3D design file of a part. The CAD file is a digital representation of the physical part, but L-PBF machines cannot directly read design files. Because of this, an engineer or designer must convert any design file into an STL file, one of the most widely used file formats for every AM process. Typically, an operator will use a separate slicer software package that will convert STL files into multiple layers, which are vertically stacked for effective part building.

Instead of converting slicer files directly to G code like many AM machines, L-PBF uses specialized build processor software that converts STL into a job file that can control the scan strategy of the machine. Scan strategy describes a broad range of functions such as laser temperature, laser path, and powder application. Scan strategy also refers to setup steps such as calculating machine movements based on factors like machine settings, material types, and part geometry. Scan strategies can affect things like layer thickness, laser temperature, and atmospheric controls of the build chamber, which can enable the creation of a repeatable build process.

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Powder Sieving System

Powder condition and particle size distribution is critical to successful printing. In most cases, an industrial sieve is used to prepare and recycle the powder for L-PBF printing. A sieve is a type of filtering device that uses mesh to separate unwanted particulate matter and large particles from powder. Many factors involving the sieve can affect an L-PBF print. For example, some L-PBF machines sieve recycled powder and store it in a closed, inert gas environment. Other machines may expose material to air during processing and sieving, which may cause oxidation and affect the print. Other factors that might affect an L-PBF build may include tears in the sieving mesh. An operator should check the mesh before conducting a print. The quantity and type of powder being used can also affect the rate at which the mesh may need to be replaced. In order to ensure proper printing, operators should avoid loading unsieved powder into an L-PBF machine.

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Machine Components

The powder hopper contains sieved powder, which is commonly pushed upward, or dosed, in very precise quantities to the powder blade, or recoating blade, which then spreads the powder onto the build platform in a very thin layer.

The build platform is a level, flat surface where the laser beam heats and fuses powder during an L-PBF print to build the part layer-by-layer. Once a layer is complete, the powder blade pushes the excess powder into the overflow container and recoats the build platform. The build piston lowers the build platform to a precise amount in preparation for the next layer.

The machine directs the laser through a lens to melt the powder on the build platform. However, a mirror and lens system is typically used to help control laser power, beam spot size, and the beam path as it travels along the build area. Typically, a window protects the laser, lens, and mirrors from particle contamination.

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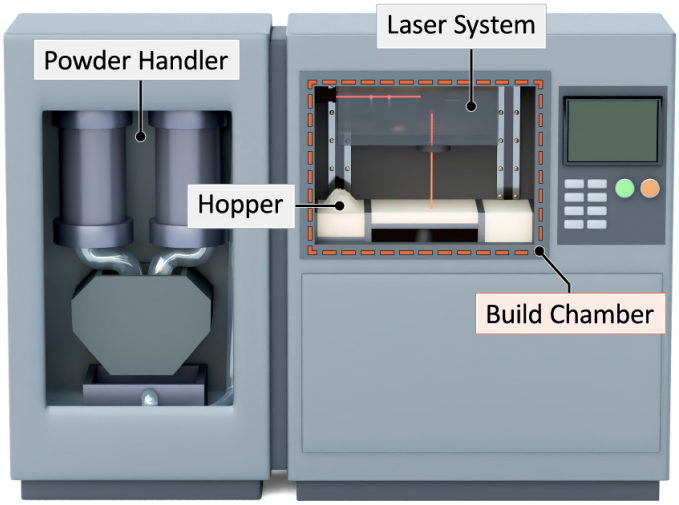
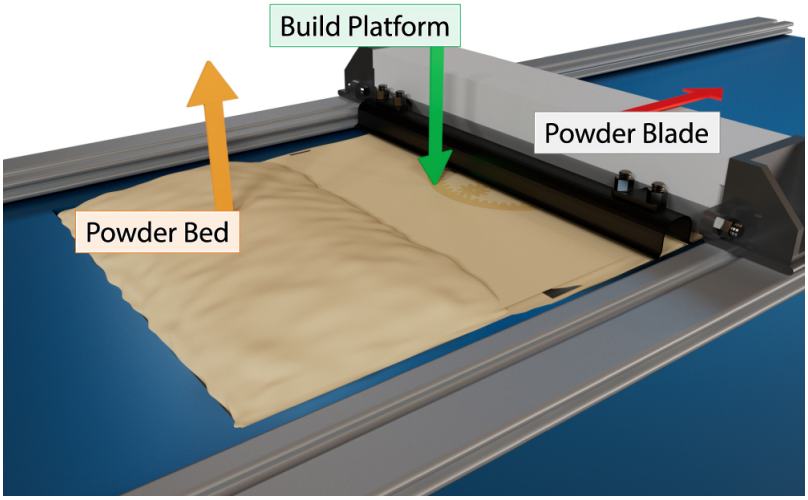
Machine Component Calibration



L-PBF Subsystems

The powder hopper, build chamber, and the laser optics areas often act like subsystems within the L-PBF machine.



Mechanical Components Inside the Build Chamber

Because L-PBF subsystems are typically self-contained, an adjustment or change with one subsystem will not affect the others. For example, an operator should verify that the build platform raises and lowers precisely. Otherwise, the platform may damage the powder blade and cause the powder to spread unevenly across the build platform. However, adjusting the build platform will typically not affect the powder hopper.

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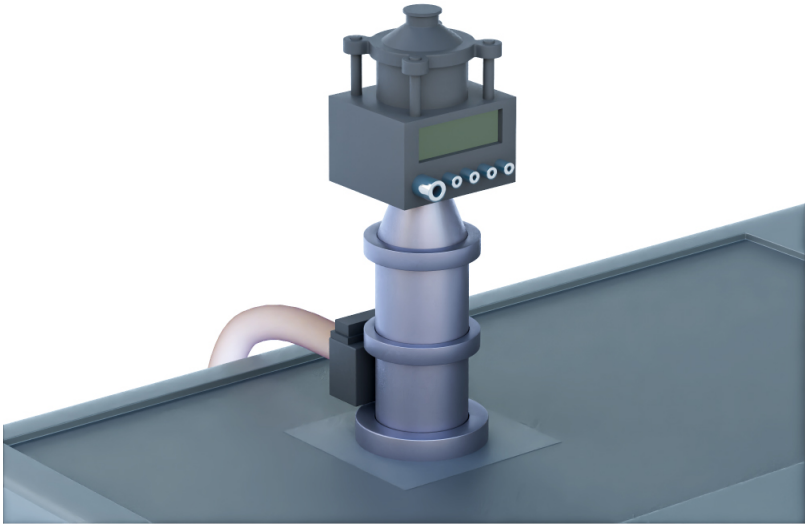
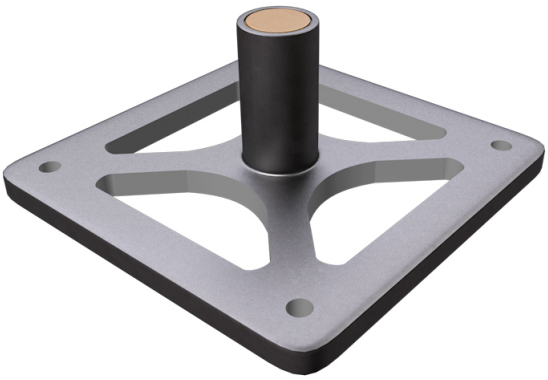


Vaccuum Feed

Loading and transferring sieved powder to the L-PBF machine can cause contamination, which can result in printing errors. Some L-PBF machines may allow an operator to manually load sieved powder directly into the hopper themselves. However, manual transference has the highest risk of cross-contamination from other powders in addition to further contamination from human hair, clothing fibers, dirt, and other substances. To avoid this, L-PBF operators may use a gravity-fed powder feed or vacuum feed to transfer the powder from the sieve to the machine in a self-contained environment.



Calorimetric Tester

An L-PBF operator can use a sensor system to check laser beam power and laser spot size before a build job. For example, calorimetric testers can check the laser beam power of most commercial LPBF systems by using a thermocouple sensor to measure the laser temperature as it travels from one point to another.

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L-PBF Component and System Review

1. Loading and transferring sieved powder to the L-PBF machine can cause contamination. **TRUE**
2. The purpose of the powder hopper is to pushe sieved powder across the build plate in very precise quantities. **FALSE**
3. Engineers work with software to create 3D design files of L-PBF parts. **TRUE**
4. L-PBF setup processes must account for variables in temperature and building strategies that might affect a part's microstructure. **TRUE**
5. In most cases, an industrial vat is used to prepare and recycle the powder for L-PBF printing. **FALSE**

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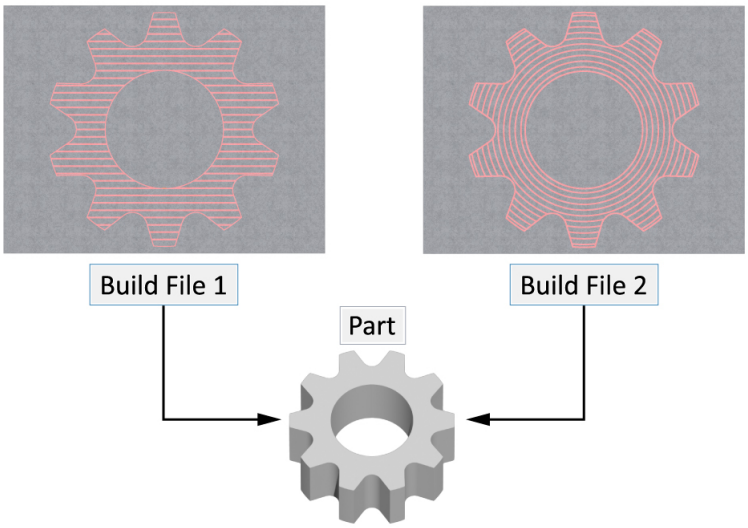
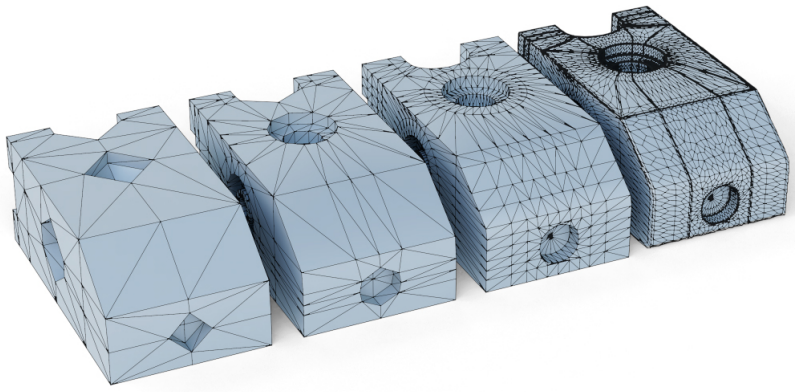
Software Configuration for Printing



It's generally possible to create the same part using several different laser build paths, which is referred to path independence. However, choosing different paths may alter a part on the microstructure level, which is affected by part geometry and the way heat is transferred from the laser. In order to remain consistent when building copies of the same part, it's best to reuse the same build file.



Building software in many L-PBF machines will convert designs into triangles, which is referred to as tessellation. The number of triangles used in a part design file determines its level of detail. The design must have enough triangles to achieve the required part dimensions. An object with too many triangles, however, may make the file too large, requiring the part build to split into different components.

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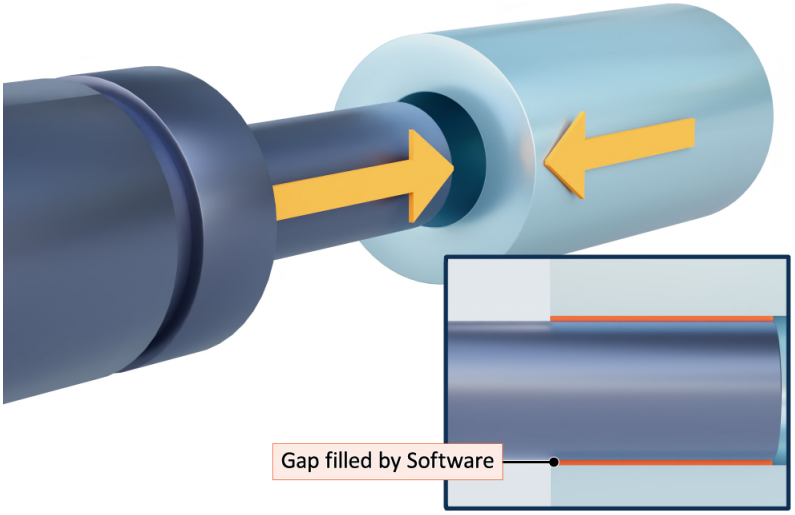
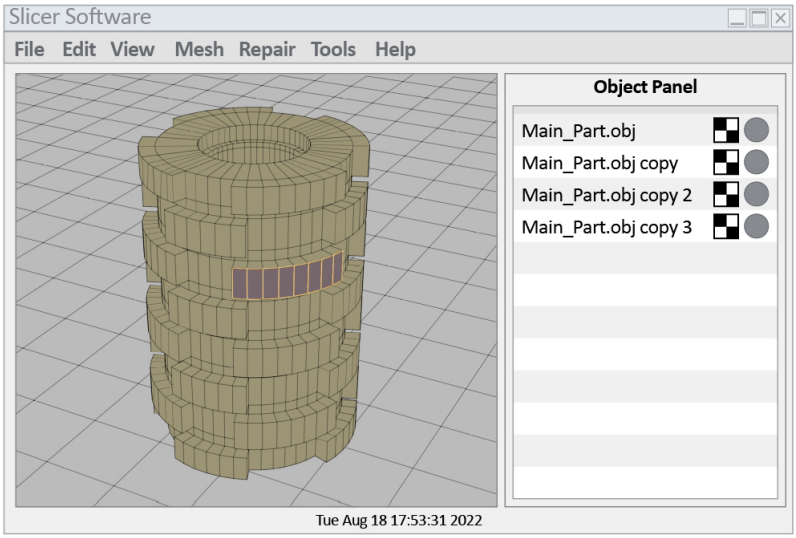


If an L-PBF printed part interfaces with another component, then layer thickness may be an issue. Using software to fill any gaps that may occur between mating parts is a common way to resolve potential print issues.



Ghost parts are another common issue that may occur during L-PBF setup. Ghost parts are typically created when an operator accidentally loads the same or similar programs two or more times into the slicer. Multiple programs create different laser paths that can still build a part that looks visually appropriate, but unreliable on the microstructure level, which can affect repeatability. Ghost parts can be avoided by doublechecking part counts before initiating a job and monitoring the L-PBF build for remelting, which should never occur.

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Mirrors and Lenses

Typically, the laser optics in an L-PBF machine include a a series of mirrors that are part of a system of mirror galvanometers, that direct the laser beam to the correct area on the build plate by measuring small electrical currents. Laser optics also include a specialized lens to control the spot size and penetration depth of the laser, which are essential for repeatability in L-PBF systems.

Mirror and lens location misalignment can cause incorrect part dimensions. During setup, L-PBF operators can check the dimensional control of the laser, which affects the geometry of the part build, and the location control, which refers to specific Cartesian coordinates on a build plate. For example, using a single laser beam to build a part that is not directly attached to the build plate reduces the risk that mirror and lens location misalignment will result in a scrapped part. A laser can have high dimensional control even if the location on the build plate is incorrect because the angle of the part build using supports does not typically affect the dimensions of a part. However, if an L-PBF machine uses multiple lasers, then mirror and lens misalignment can cause defects between the build layers. If the laser location on the build plate is incorrect, it typically causes part layers to overlap. This is commonly referred to as stitching. In addition, an L-PBF part may experience high rates of porosity in areas where stitching has occurred. Because many geometric defects happen internally within a part and are difficult to discover with nondestructive testing (NDT) methods, calibrating mirrors and lenses before printing can help avoid internal part defects.

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Build Surface Setup

In order to protect the build platform, most L-PBF operators will use a build plate, or start plate, to build the actual part.

Build plates for L-PBF machines are typically thin, flat, and made of metal. A build plate needs to be completely flat and level or it may affect the geometry of the part build. A build plate with imperfections may also affect the uniformity of how the powder is spread across the build area. L-PBF parts typically need to be cut off of the build plate after printing, which can cause imperfections and uneven areas. Because of this, operators should resurface build plates before reusing.

An operator should also consider strength and stiffness when selecting a suitable build plate. If a build plate is too weak, it may warp or deform during the print and affect the print geometry. One way to lessen the effects of imperfections in the build plate is by using support structures, which help reinforce the part as it is being built. If the first few layers of a part build are support material, it may significantly reduce the impact of an uneven build surface.

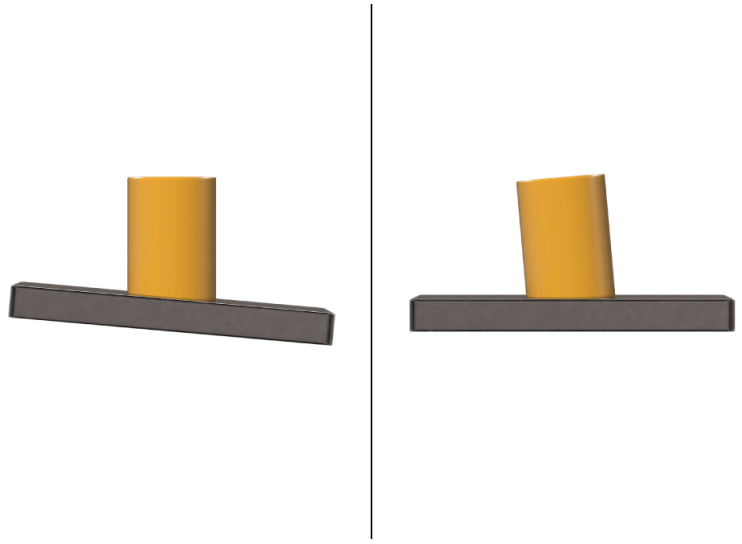

Figure 1: Uneven Build Plates May Affect Part Geometry

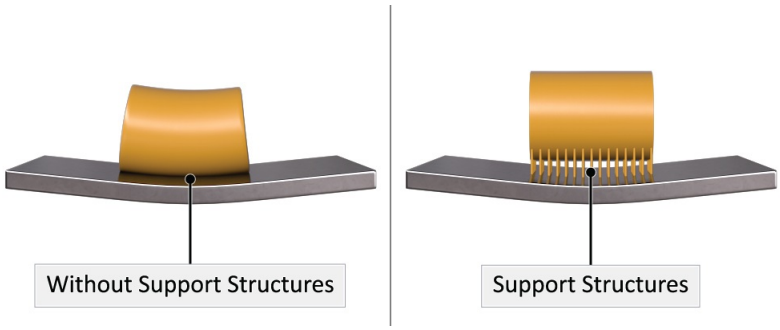

Figure 2: Support Structures Lessen Effects of Build Plate Imperfections

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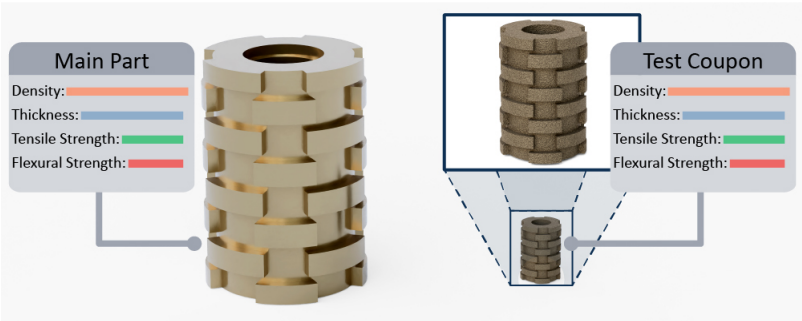
Test Prints and Coupons



No standard currently exists for laser position control due in large part to the variation that exists among different types of L-PBF machines. One common way to determine laser position involves test printing, or burning a pattern into a piece of laser paper directly onto a new build plate. Measuring the pattern establishes a baseline for laser alignment. However, measuring perpendicular laser lines using this method may prove problematic and require a special testing system like a circular aperture, which can measure the beam in a distinct spot during setup. However, perpendicular laser testing in L-PBF has difficulty tracking perpendicular laser spot size as it moves. In addition, burn patterns during testing can have microscopic variations, which can affect the microstructures of a part and may make small misalignments difficult to identify.

Another method of L-PBF testing can involve adding a test coupon, which can be built separately during a part build or attached as a protrusion to an end-use part and removed during post-processing. Test coupons can act as a sample part or piece of material made for the purpose of testing and inspection. Test coupons typically have exactly the same variables as outlined in the part specification. An inspector can analyze a test coupon for things like porosity, geometric accuracy, structural integrity, and corrosion rates to determine if they change over time. Test coupons may also enable operators to determine how gases in the build environment have affected previous builds over time, which can inform setup calibrations for new L-PBF parts.

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Lesson: 11/11

L-PBF Print Setup Review

- **Test Coupon**: A sample part or piece of material made for the purpose of inspection.
- **Build Plate**: The removable platform that supports an in-process AM part during a build.
- **Path Independence**: The ability to print the same L-PBF part using different laser paths.
- **Galvanometer**: Directs the laser beam to the correct area on the build plate.
- **Support Structure**: A reinforcing component used to hold the weight of an additively manufactured part.
- **Porosity**: An empty space between the grains and bond in an L-PBF made part.

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Class Vocabulary

3D Three-dimensional. Having height, width, and depth. 3D models for AM are created with computer-aided design software.

Additive Manufacturing AM. The process of joining or solidifying materials to make an object based on a three-dimensional (3D) computer model. Additive manufacturing methods typically build up layers of material to create an object.

AM Additive manufacturing. The process of joining or solidifying materials to make an object based on a three-dimensional (3D) computer model. AM methods typically build up layers of material to create an object.

Build Chamber The area contained within an additive manufacturing (AM) machine where the part is actually built. Many build chambers are enclosed to improve the safety of the manufacturing operation.

Build Plate The removable platform in an additive manufacturing system that supports the in-process part. A build plate is commonly used in metal powder bed fusion operations.

Build Platform The flat surface on which a part is additively manufactured. The build platform can either be a permanent machine surface from which parts are removed or a surface that can be removed from a machine once the build is complete.

Build Processor A type of software used in powder bed fusion (PBF) or other additive manufacturing (AM) machines that converts STL files into job files. Build processor software is usually specific to each machine.

Calibration The comparison and adjustment of a device with unknown accuracy to a device with a known, accurate standard. Calibration eliminates any variation in the device being checked.

Calorimetric The measurement of the amount of heat an object or process releases. Calorimetrics can be used in L-PBF to monitor and measure melting behavior.

Cartesian Coordinates A numerical system that describes the location of an object by expressing its distance from a fixed position along three linear axes. Cartesian coordinates are used to direct machine tool movements.

Ceramics A hard, brittle material that can withstand high temperatures and resist corrosion. Ceramic parts can be created through additive manufacturing (AM) processes such as binder jetting and selective laser sintering (SLS).

Circular Aperture A circular hole in an object where light passes through. Circular apertures are very important in optical devices.

Computer-Aided Design CAD. A computer system used to design a model of a product. CAD models are converted into STL format for use by additive manufacturing machines.

Dimensional Control The ability of a machine or device to meet the size and specifications of the intended design. Dimensional control is essential for maintaining a repeatable target size when manufacturing a part.

End-Use Designed to be used directly by a consumer or directly in another manufactured product. End-use products created by additive manufacturing (AM) include medical implants, custom dental devices, and camera equipment.

G Code A programming language that uses letters and numbers to form commands for machine tools. G code programs are used to direct tool movements in additive manufacturing, CNC machining, and hybrid machining.

Galvanometers The mirror control for a laser on an L-PBF machine. Galvanometers direct the laser beam to the correct area on the build plate by measuring small electrical currents.

Ghost Parts Created when an operator accidentally loads the same or similar programs two or more times into the slicer. Ghost parts can affect the microstructure of a part print even if it appears visually correct.

Gravity-Fed Powder Feed An inclined tube or similarly sloped surface that enables an operator to load granular material. Gravity-fed powder feeds typically require an operator to manually scoop the material into the machine, which creates a higher risk of contamination compared to vacuum systems.

Hardness The ability of a material to resist scratching, abrasion, indentation, or cutting. Hardness is generally measured by the depth or area of penetration under a fixed load using a diamond indenter.

Hopper Containers that hold raw materials. Hoppers serve as the entryway for powder into an L-PBF machine.

Interfaces The interaction point or surface between two or more separate parts that work together. Interfaces are the connection points between two parts.

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Laser Beam An intense beam of light that can be precisely aimed and controlled. Laser beams are used in laser powder bed fusion (L-PBF) and other additive manufacturing (AM) processes.

Laser Optics Special types of lenses in an additive manufacturing machine that focus the beam to a pinpoint. Laser optics can vary in strength and types.

Laser Powder Bed Fusion L-PBF. An additive manufacturing (AM) method that uses a laser to fuse or melt together layers of powdered polymer, metal, ceramic, or other material. Laser powder bed fusion processes include selective laser sintering (SLS) and selective laser melting (SLM).

Lens A piece of glass shaped with two opposite curved surfaces that is used in optical instruments to form an image by focusing light. A lens can be different sizes for different resolutions.

Location Control The ability of a machine or device to meet the placement specifications of design features on a building platform or stage. Location control is crucial in L-PBF when the part attaches directly to the build plate.

L-PBF Laser powder bed fusion. An additive manufacturing (AM) method that uses a laser to fuse or melt together layers of powdered polymer, metal, ceramic, or other material. L-PBF processes include selective laser sintering (SLS) and selective laser melting (SLM).

Mating Parts The object that a workpiece has been manufactured to fit. Common mating parts for workpiece holes include screws and bolts.

Microns A metric measurement equal to 0.000001 of a meter, or approximately 0.000039 of an inch. Micron is a shorter term for micrometer.

Microstructure The shape and alignment of the microscopic components of a metal. A material's microstructure often determines its hardness, toughness, and other properties.

Nondestructive Testing A category of inspection processes that evaluates properties and performance using methods that do not damage or permanently alter the equipment. Nondestructive testing methods include visual inspection, liquid penetrant testing, magnetic particle testing, eddy current testing, ultrasonic testing, and radiographic testing.

Path Independence Not following defined parameters. Path independence in L-PBF can refer to building the same part using several different laser build paths.

Penetration Depth The depth to which the laser heat can melt base metals. Penetration depth partially depends on the strength of the laser.

Perpendicular An intersection of two lines or objects at a right angle. Perpendicular lines are used to determine several aspects of tool geometry, including rake angles.

Piston A small cylinder that moves up and down inside a hollow cylinder. Pistons may be used in a laser powder bed fusion (L-PBF) system to transfer powder.

Polymers A lightweight material that generally has high corrosion resistance, a high strength-to-weight ratio, and a low melting point. Polymers, which make up all plastics, are some of the most common manufacturing materials.

Porosity An empty space between the grains and bond in an L-PBF made part. Porosity is often determined by the number of small voids within an L-PBF part.

Post-Processing A group of procedures that are used to clean, improve, or otherwise finish a part for use by a manufacturer or consumer. Post-processing for additively manufactured parts includes abrasive finishing, heat treatment, and painting.

Powder Blade A device that slides forwards and backwards over the surface of a powdered material in a powder bed. Powder blades are used in laser powder bed fusion (L-PBF) systems to smooth powdered material layers over a part or build platform.

Recoating Blade A device that slides forwards and backwards over the surface of a powdered material in a powder bed. Recoating blades, also referred to as powder blades, are used in laser powder bed fusion (L-PBF) systems to smooth powdered material layers over top a part or build platform.

Remelting The temperature necessary to change a solid to a liquid more than once. Remelting during L-PBF operations can be problematic.

Repeatability The ability of a machine or process to continually deliver consistent and uniform results. Repeatability is crucial for efficiently producing parts that meet quality requirements.

Scan Strategy A term used with L-PBF systems to describe laser path and laser parameters. Scan strategy can describe a broad range of L-PBF functions like laser speed and spot size as well as the type of pattern used to create an object.

Scrap Unusable material produced during a manufacturing process. Scrap is a waste product of manufacturing that can increase overall production costs.

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Sensor A device that detects a physical stimulus and turns it into a signal that can be measured and recorded. Sensors make it possible for LPBF machines to communicate digitally with other machines.

Sieve A type of filtering device used to separate unwanted particulate matter and large particles from powder. Sieves are often used when working with metal powder for additive manufacturing.

Slicer A computer program that divides an STL file into layers for additive manufacturing (AM). Slicers can either be separate computer programs or part of the machine interface.

Software Coded instructions or programs that control computer hardware functions and operations. Software is used as a platform to design additively manufactured parts.

Spot Size The point at which a laser beam meets after reflection or refraction. Spot size, or focal point, is often created by a lens or a series of lenses on an L-PBF machine.

Stitching Unintentional overlapping between two interlocking parts in additive manufacturing. Stitching in L-PBF may be caused by misaligned lasers.

STL An additive manufacturing (AM) compatible file format that represents 3D models as a series of interconnected triangles. STL files are sometimes referred to as stereolithography files, standard tessellation language files, or standard transform language files.

Support Structures A reinforcing component used to hold the weight of an additively manufactured part as it is being constructed. Support structures are removed from the part once the build is complete.

Tessellation The arrangement of shapes together. Tessellation in L-PBF refers to the conversion of a surface model to an equivalent triangle or polygon representation.

Test Coupon A sample part or piece of material made for the purpose of testing and inspection. Test coupons have exactly the same variables as outlined in the part specification.

Thermocouple A temperature sensor that measures heat at a hot junction and outputs voltage at a cold junction. Thermocouples output voltage that is proportional to the measured temperature.

Toughness The ability of a material to withstand forces or sudden impacts that attempt to break it. Toughness allows materials to withstand sudden stress.

Vacuum Feed An airtight tube or duct used to move granular material by suction. Vacuum feeds are used in LPBF machines to avoid contamination of powder material when transferring from sieving system to powder bed.

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